ENSEMBLE HEALTH PARTNERS
AI / ML Team

Take-Home Technical Assessment

Binary Classification · Model Evaluation · GenAI Explanations

Time estimate	4–6 hours. Focused, well-explained work is more valuable than complex work.
Submission	GitHub link + Predictions + PDF write-up (2–3 pages).
Deadline	Within 5 business days of receiving this document.
Questions	Email your recruiting coordinator — we reply within one business day.

No healthcare background needed
Everything you need to understand the context is explained below.
The skills being tested — problem framing, classification, evaluation, and GenAI — are general.

The Problem

When a hospital submits a bill to an insurance company, the insurer either pays it or denies it. Denials create rework — the hospital must investigate, correct, and resubmit, which costs time and money.

Many denials are predictable from information available before the claim is submitted: missing authorization, incomplete documentation, eligibility gaps, network issues. A front-end review team uses this signal to catch risky claims before they go out — but they can only manually review a fraction of the daily volume.

Your task

Part 1 — Build a binary classifier that predicts whether a claim will be denied. Train and evaluate on claims_history.csv. Score current_claims.csv.

Part 2 — For the top 10 highest-risk current claims, use an LLM to generate a short, plain-English explanation an analyst can read and act on in seconds.

The review team can inspect only the top 25% of claims by risk score before submission.
That constraint should drive your metric choices.

The Data

Two synthetic CSV files — no real patient information is included.

claims_history.csv	3,200 historical claims with known outcomes. Includes a split column (train / validation / test) — use it as-is.

current_claims.csv	500 current claims. No outcome known. Score these and generate explanations for the top 10.

Column Reference — claims_history.csv

Column	Type	What it means
claim_id	text	Unique ID. Do not use as a feature.
payer_id	text	Which insurance company (P001–P010).
payer_type	text	Commercial, BCBS, Medicare Advantage, or Medicaid MCO.
visit_type	text	Inpatient, Outpatient, Emergency, or Observation.
total_billed	number	Total amount the hospital charged.
expected_payment	number	Amount the hospital was contractually entitled to receive.
num_procedures	number	Number of procedure codes on the claim.
num_diagnoses	number	Number of diagnosis codes on the claim.
prior_auth_required	0 / 1	Whether the insurer requires advance approval for this service.
has_prior_auth	0 / 1	Whether that approval is actually on file.
is_in_network	0 / 1	Whether the provider is in the insurer's network.
days_to_submit	number	Days between care being provided and the claim being submitted.
missing_documentation_flag	0 / 1	Whether required supporting documents appear to be missing.
eligibility_verified	0 / 1	Whether the patient's insurance coverage was confirmed beforehand.
referral_required	0 / 1	Whether the insurer requires a referral for this service.
referral_present	0 / 1	Whether that referral is on file.
split	text	train / validation / test. Use this — do not create your own split.
service_month	text	Month of service (YYYY-MM).
is_denied	0 / 1	TARGET. Do not use as a model input.
denial_reason	text	Plain-English reason for denial. Only exists after a denial — do not use as a model input. You may use it to analyze errors after the fact.

> **Watch out for target leakage**
> is_denied is your prediction target — it should not be used for training.
> denial_reason only appears after a denial has occurred.
> Using either as a model input will make training look great but fail on real data.

current_claims.csv has the same columns minus split, is_denied, and denial_reason — exactly the information available before submission.

Deliverables

Score the current claims

Apply your final model to current_claims.csv and produce a file called predictions_current_claims.csv sorted from highest to lowest denial_probability.

claim_id	From current_claims.csv.
denial_probability	Predicted probability of denial (0.0–1.0).
predicted_denial	0 or 1 at your chosen threshold.
risk_tier	High / Medium / Low. Define your thresholds in the README.
top_risk_factors	Two or three features most responsible for this claim's score.
explanation	Short plain-English explanation — generated in next step.

Generate explanations with a language model

For the top 10 highest-risk claims in current_claims.csv, use an LLM to write a short explanation an analyst can read and act on in seconds.

You may use any LLM API (OpenAI, Claude, Gemini, a local model). If you do not have API access, write the prompts and include two or three manually drafted example outputs — we are evaluating your prompt design, not just whether you made an API call.

Each explanation should:

- Be grounded only in the claim's actual field values and the model's risk drivers. Do not invent facts.
- Use plain language — no jargon the analyst would need to look up.
- Include one specific recommended action.
- Acknowledge that this is a risk estimate, not a guarantee of denial.
- Be 2–3 sentences long.

Include your prompt template in your submission. Show what happens when you give it a low-risk claim — does it behave sensibly?

PDF Write-Up (2–3 pages)

- How you framed the problem — which errors matter more and why.
- Two or three data findings worth sharing with a non-technical manager.
- What you built, what baselines you compared against, and your threshold choice.
- Test set metrics and denial capture at top 25%.
- Your LLM prompt, one example explanation, and one honest limitation.
- What you would improve with more time.

Ensemble Health Partners | Confidential — For Candidate Use Only | Do Not Distribute